



# CPSC 100

# Computational Thinking

## Artificial Intelligence

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Department of Computer Science  
University of British Columbia

# Agenda

- Artificial Intelligence
- Machine Learning
- Algorithmic bias
- Bias and Fairness
- Preview of Data Mining

# Course Admin

# Lab 8 due date extension

## Task 6 - KMeans Introduction

[View...](#)

Instructor's Note: Please note that content from this task has not yet been covered in class; the plan is for the TAs to introduce this technique in Labs during Week 10 (the week of March 9), and for us to start talking about this in class on March 23rd. The due date for Lab 8 has been adjusted to account for this.

## Task 6 - Introduction to KMeans Algorithm

In this task you will dive a bit deeper into the KMeans algorithm for clustering. Remember that "clustering" is an "unsupervised (un-labeled data) learning" technique that groups data points together based on their similarity to other neighbouring data points and dissimilarity to data points that are further away.

### KMeans Algorithm



# Learning Goals

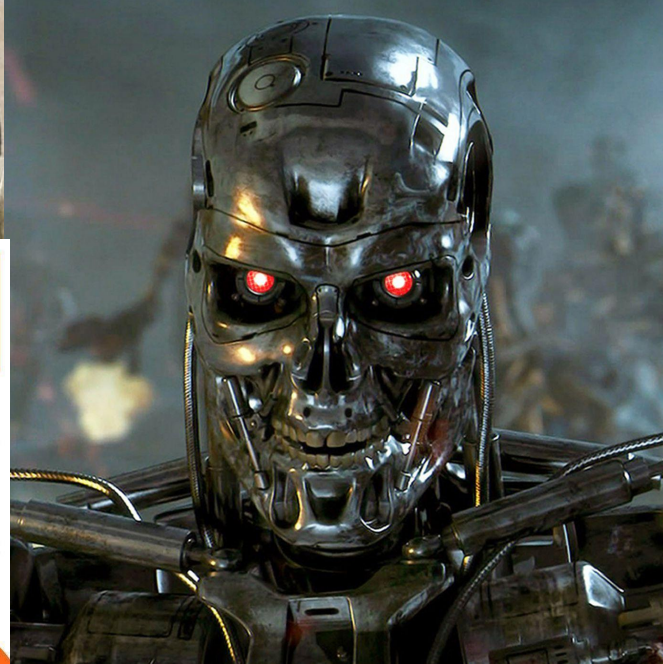
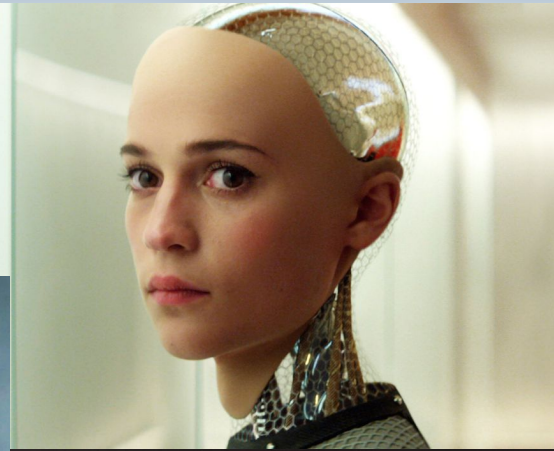
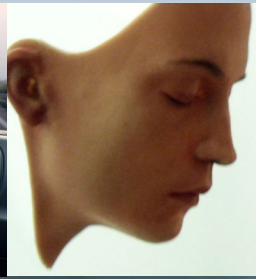
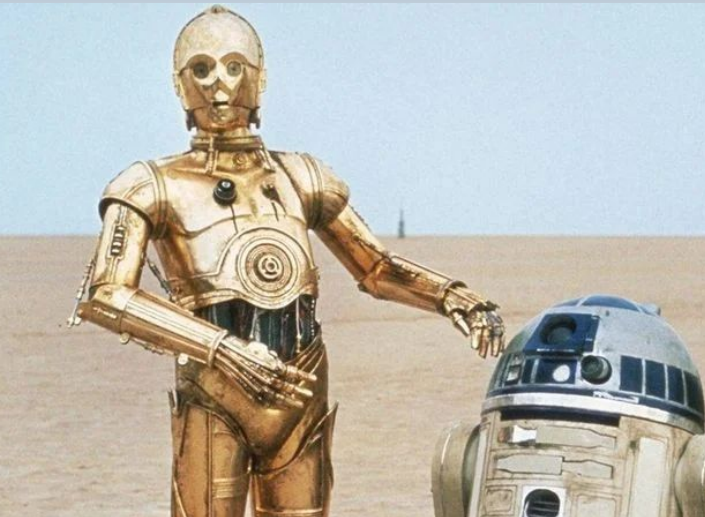
After this week's lecture, you should be able to:

- Explain the term “Artificial Intelligence” and be able to determine which methods are fall within its umbrella
- Explain what “Machine Learning” is conceptually
- Determine whether particular tasks are well-suited for machine learning or not
- Describe the difference between Supervised and Unsupervised Machine Learning
- List and describe common machine learning types
- Assess whether analyses have algorithmic bias

# Artificial Intelligence

Slides courtesy of Dr. Kevin Leyton-Brown, Director of Center for AI  
Decision-making and Action (CAIDA) and CIFAR Chair

# What AI Means to Most People



# But most modern AI systems look like...



# What is Artificial Intelligence?

AI is the study, design, and development of computational processes that solve problems that **previously required human intelligence**

Not all of computer science is focused on AI.

But, it's hard to think of a field in CS that isn't **contributing fundamentally** to it

The “**AI Paradox**”: once we become familiar with a technology, we stop considering it AI

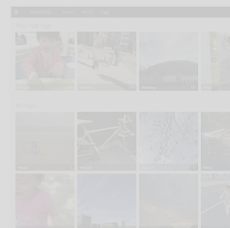
IBM computer Watson wins  
clash

DeepFace: Challenge the  
Yanis Tai

## Microsoft, Google Image Recognition

Deep learning algorithms compete

PORTLAND, Ore. — First computers hurdle is image recognition — surely, too. Now Microsoft has programmed the competition is fierce, with the first the 2015 championship on December they have one-upped humans too. For benchmark of 5.1% errors with a 4.94 Microsoft by 0.04%.



ImageNet, with hundreds of object categories, has been the main competition since 2010 with about 300,000 images. The goal is to take the crown from the best human score. All the contest algorithms, which are all derived from various versions of a human brain works to varying degrees. Most of the contest algorithms to meet detail — in the context of some examples.

## OpenAI's Dota 2 AI steamrolls world champion esports team with back-to-back victories

The Int  
By Nick S

HEALTH + BEHAVIOR

Artificial intelligence performs as well as experienced radiologists



## New AI Model Exceeds Human Performance at Question Answering

Microsoft's new AI translates Chinese-to-English as well as a human translator

By Michael Hicks March 14, 2018 World Of Tech

It's all in how you teach the AI



Denise Hea

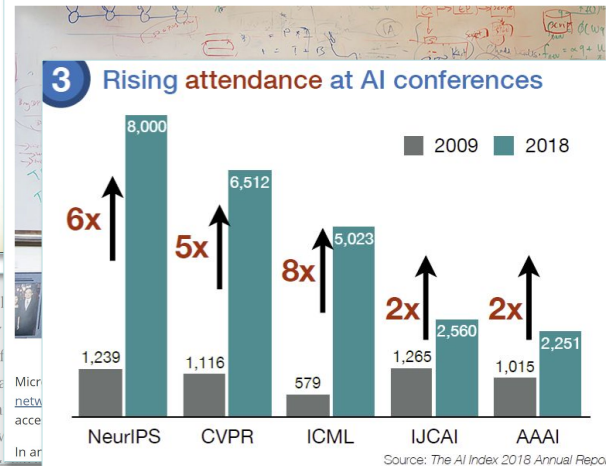
FINDINGS

UCLA research diagnose prostate cancer by evaluating 10 years of experience

BACKGR

Radiologists practicing on accurately es highly special

METHOD



On May 11, 1997, IBM's Deep Blue® beat the world chess champion, Garry Kasparov, in a six-game match. The victory was the classic the contest, but the machine's triumph was a

1995 - 1998

2011

2014

2015

2016

2017

2018+

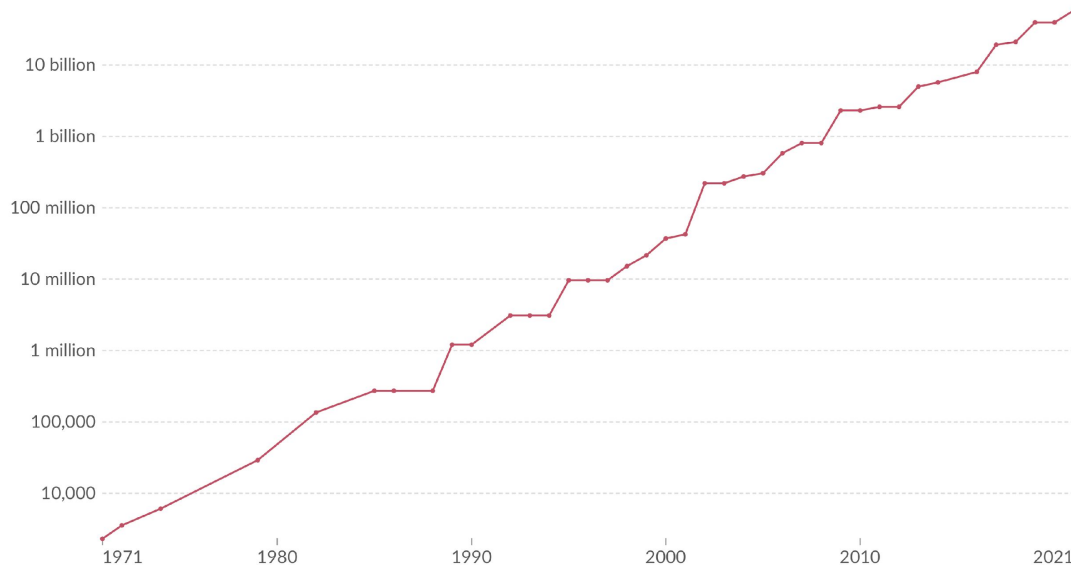
# Why is it happening?

1. Scientific/mathematical **breakthroughs**, especially in machine learning
2. Growth in raw **computing power**

## Moore's law: The number of transistors per microprocessor

Our World  
in Data

Moore's law is the observation that the number of transistors in an integrated circuit doubles about every two years, thanks to improvements in production. It was first described by Gordon E. Moore, the co-founder of Intel, in 1965.



Data source: Karl Rupp, Microprocessor Trend Data (2022)

OurWorldInData.org/technological-change | CC BY

# Current State of the Art

- **Predictive modeling**  
limited mostly by quality of training data
- **Image recognition, face recognition**  
becoming extremely reliable
- **Image generation** almost there
- **Language understanding & generation**  
maturing extremely quickly. Watch this space
- **Fueling scientific breakthroughs**  
from protein folding to astronomy
- **Robotics** remains a hard problem, but it's working  
well in some niche applications



Prompt: Here's a short rhyming poem by Dr. Seuss about Elon Musk's company, Neuralink which is building brain implants to achieve symbiosis with AI. Neuralink's mission statement is "If you Can't Beat'em, join'em". The poem has a structure typical of Dr. Seuss stories with simple alternate line rhymes. In the poem Elon explains why we need to merge with the AI to save humanity.

Neuralink, by Dr. Seuss

-----  
Response:

I have a friend, Elon Musk,  
Who wants to join our brains to AI.  
He thinks that humans can't compete  
So he's gonna use AI to cheat.  
They want to build a neural-link.  
To link us to the AIs  
And save humanity from its demise.  
They want to put an AI in your brain.  
They want to use your neurons as a frame.  
They want to save humanity,  
If you can't beat'em, join'em

<https://www.gpt-3.com> (gpt-3)

# Machine Learning

- Narrow: building **predictive models** from data
- Broad: a new approach to computer programming, where computers **learn from experience** rather than being told what to do
- Where's it **headed**?
  - More and more data collection
    - sensors
    - behavior tracking
  - Image, text, speech understanding
  - Data mining/personalization/profiling
  - Self programming/optimization



# Activity: Spot the Fakes

- Do the quiz once individually (3 minutes)
- How'd you do?
- Do the quiz together in a group of 3 or 5



## Real or Not ?

The growing quality in AI images makes them harder to spot.  
**Can you tell if this image is real or AI generated?**

Start Game

# Activity: Spot the Fakes

## Strategies for spotting the Fakes

- Continuity e.g Clothing: follow seams, holding guns wrong, knives, body structure
- Hands
- Blurriness
- Certain objects are hard to render (cameras, brands, logos)
- Lighting & Reflections
- Blatant errors (smiling, extra teeth)
- Physics doesn't add up (things that are wet)

# Machine Learning

# Machine Learning

## A different way to solve problems

Machine learning uses computer programs to model data. It can be used to extract hidden patterns, make predictions in new situation, or generate novel content.

A field of study that gives computers the ability to learn without being explicitly programmed.

– Arthur Samuel (1959)

# Machine Learning

## Traditional programming vs. ML

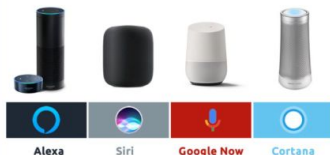
- Imagine writing a Python program for spam identification, i.e., whether a text message or an email is spam or non-spam.
- Traditional programming
  - Come up with rules using human understanding of spam messages.
  - Time consuming and hard to come up with robust set of rules.
- Machine learning
  - Collect large amount of data of spam and non-spam emails and let the machine learning algorithm figure out rules.

# Machine Learning

## Prevalence of ML

Let's look at some examples.

### Voice assistants



### Google news

- Armed man who broke into Trudeau residence charged with threatening to kill or injure PM  
The Guardian - 1 hour ago
- Corey Hurren, alleged Rideau Hall intruder, threatened Trudeau: RCMP officer  
Global News - 4 hours ago
  - Corey Hurren had multiple firearms, uttered threat against Trudeau, court documents allege  
CBC.ca - 2 hours ago
  - Man arrested near Rideau Hall had several weapons, threatened PM Trudeau: RCMP  
CTV News - 22 minutes ago

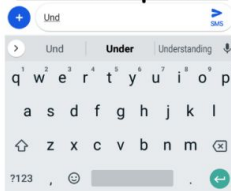
### Recommendation systems



### Face recognition



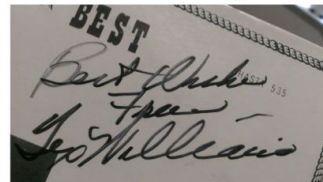
### Auto-completion



### Stock market prediction



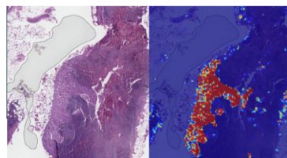
### Character recognition



### Self-driving car



### Cancer diagnosis



### Drug discovery



### AlphaGo



# Machine Learning

## Activity: For what type of problems ML is appropriate? (~5 mins)

Discuss with your neighbour for which of the following problems you would use machine learning

- Finding a list of prime numbers up to a limit
- Given an image, automatically identifying and labeling objects in the image
- Finding the distance between two nodes in a graph

# Machine Learning

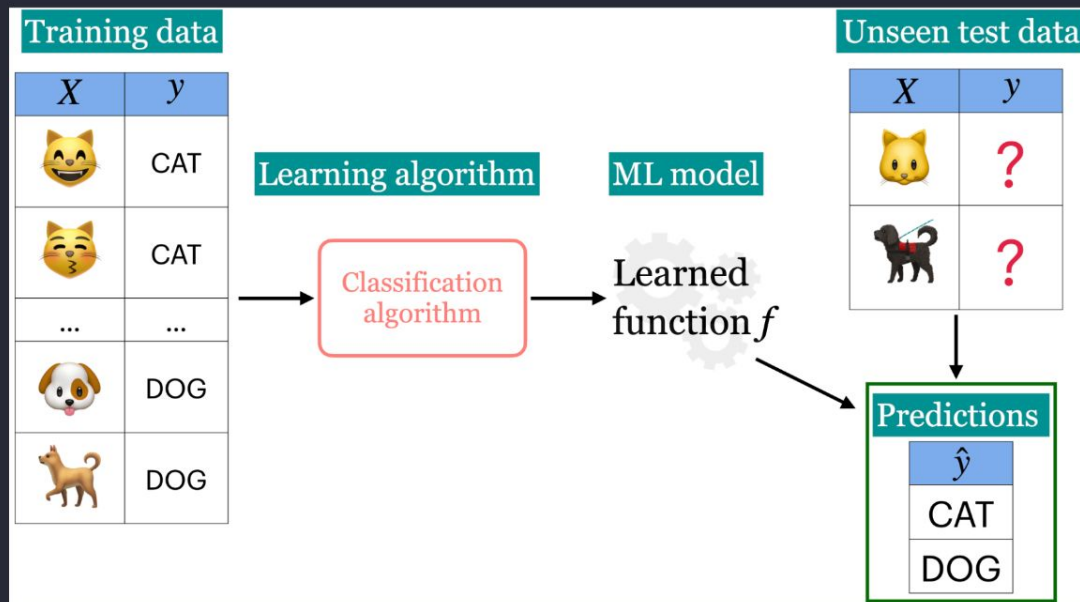
## Types of machine learning

Here are some typical learning problems.

- **Supervised learning** (Gmail spam filtering)
  - Training a model from input data and its corresponding targets to predict targets for new examples.
- Unsupervised learning (Google News)
  - Training a model to find patterns in a dataset, typically an unlabeled dataset.
- Reinforcement learning (AlphaGo)
  - A family of algorithms for finding suitable actions to take in a given situation in order to maximize a reward.
- Recommendation systems (Amazon item recommendation system)
  - Predict the “rating” or “preference” a user would give to an item.

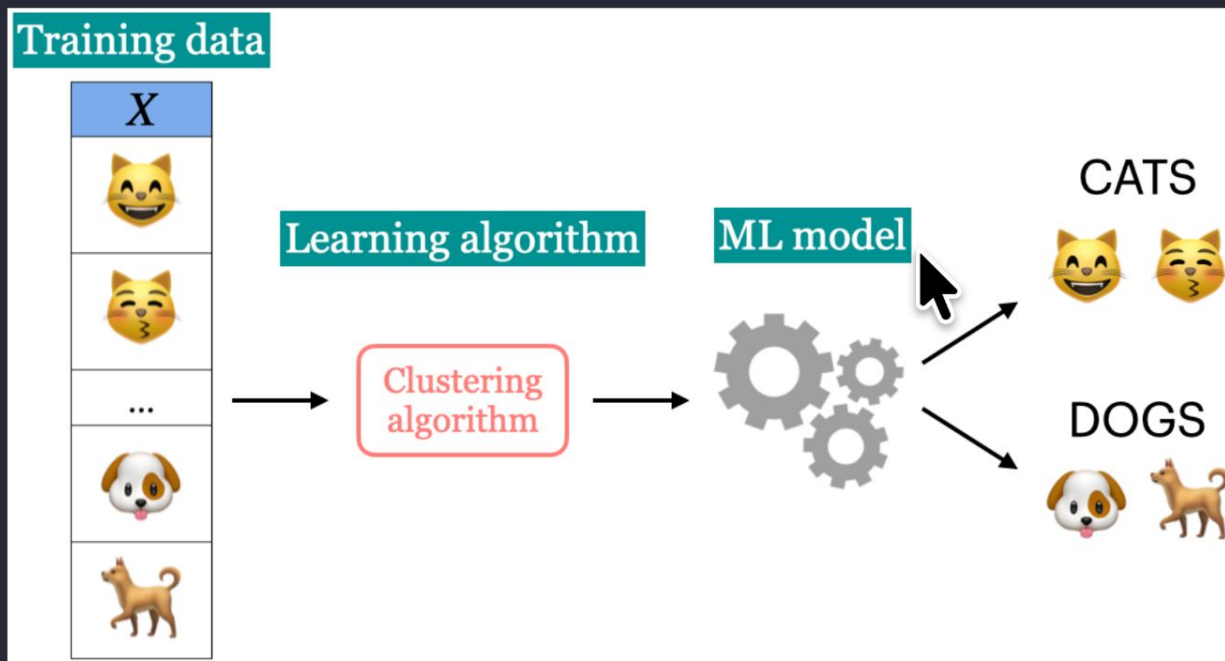
# What is supervised learning?

- Training data comprises a set of observations ( $X$ ) and their corresponding targets ( $y$ ).
- We wish to find a model function  $f$  that relates  $X$  to  $y$ .
- We use the model function to predict targets of new examples.



# Unsupervised learning

- Training data consists of observations ( $X$ ) without any corresponding targets.
- Unsupervised learning could be used to group similar things together in  $X$  or to find underlying structure in the data.



# Wrap-up

# Bias and Fairness

# Human bias

Bias in people refers to our tendency to take quick decisions based on little information

Published online 11 April 2011 | Nature | doi:10.1038/news.2011.227

**News**

## Hungry judges dispense rough justice

When they need a break, decision-makers gravitate towards the easy option.

Zoë Corbyn

*Journal of Economic Perspectives—Volume 12, Number 2—Spring 1998—Pages 41–62*

## Evidence on Discrimination in Mortgage Lending

Helen F. Ladd

## Science faculty's subtle gender biases favor male students

Corinne A. Moss-Racusin<sup>1,2</sup>, John F. Dovidio<sup>3</sup>, Victoria L. Brescoll<sup>4</sup>, Mark A. Graham<sup>1,2</sup>, and Jo Handelsman<sup>1,2</sup>

<sup>1</sup>Department of Molecular, Cellular, and Developmental Biology, <sup>2</sup>Department of Psychology, <sup>3</sup>School of Management, and <sup>4</sup>Department of Psychiatry, Yale University, New Haven, CT 06520

Edited by Shirley Tilghman, Princeton University, Princeton, NJ, and approved August 21, 2012 (received for review July 2, 2012)

**The Observer**  
Stop and search

## Racial bias in police stop and search getting worse, report reveals

Despite reforms, black people are nine times more likely than white people to be checked for drugs

Mark Townsend  
Home Affairs  
Editor

4 comments  
Sat 10 Jul 2010 20:23 BST

1716  
The article is one of 5 featured

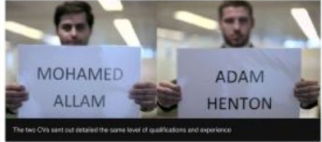


## Is it easier to get a job if you're Adam or Mohamed?

By Zack Abramson and Clara Mancuso  
BBC Inside Out

6 February 2017

f t e g+ Share



The two CVs sent out detailed the same level of qualifications and experience

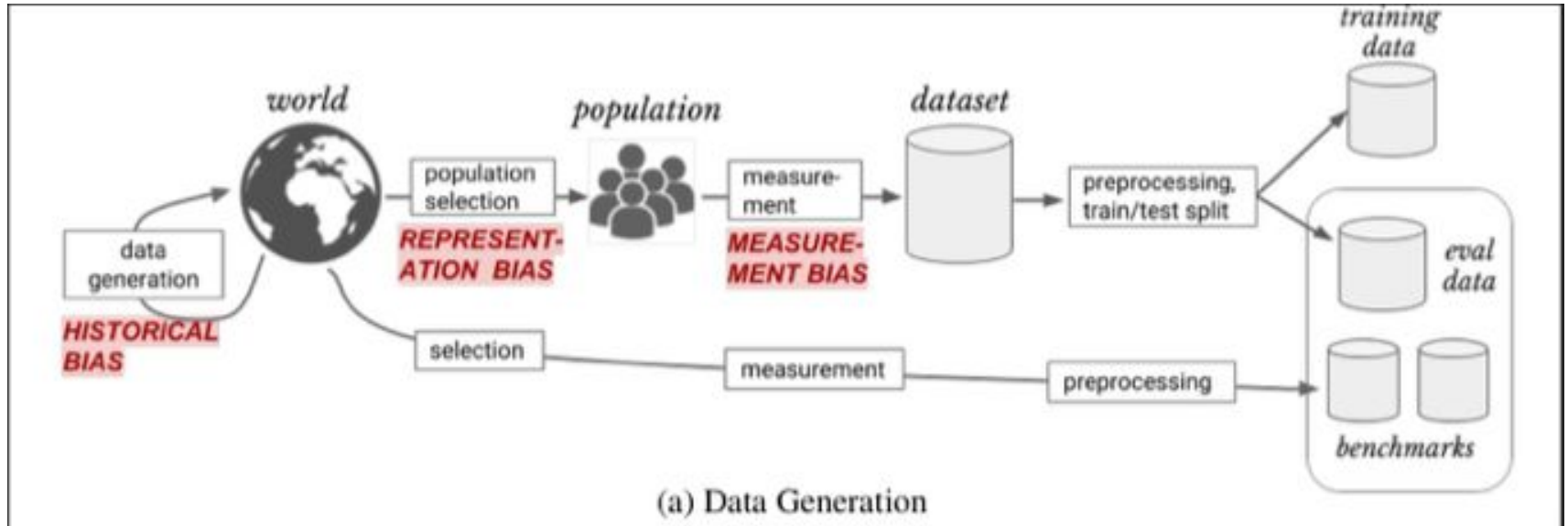
A job seeker with an English-sounding name was offered three times the number of interviews than an applicant with a Muslim-sounding name, BBC test found.

# Why worry about bias in algorithms

Decisions made by a ML algorithm are:

- Cheap
- Scalable
- Automated
- Self-reinforcing
- Seemingly objective
- Often lacking appeals processes
- Not just predicting but also causing the future

# Sources of bias in ML algorithms



# Representation bias

**Representation bias** arises while defining and sampling a development population. It occurs when the development population under-represents, and subsequently fails to generalize well, for some part of the use population. 1

1. **The sampling methods only reach a portion of the population.** For example, datasets collected through smartphone apps can under-represent lower-income or older groups, who are less likely to own smartphones. Similarly, medical data for a particular condition may be available only for the population of patients who were considered serious enough to bring in for further screening.
2. **The population of interest has changed or is distinct from the population used during model training.** Data that is representative of Boston, for example, may not be representative if used to analyze the population of Indianapolis. Similarly, data representative of Boston 30 years ago will likely not reflect today's population.

# Measurement bias

**Measurement Bias** arises when choosing and measuring features and labels to use; these are often proxies for the desired quantities. The chosen set of features and labels may leave out important factors or introduce group- or input-dependent noise that leads to differential performance.

## 3. **The defined classification task is an oversimplification.**

In order to build a supervised ML model, some label to predict must be chosen. Reducing a decision to a single attribute can create a biased proxy label because it only captures a particular aspect of what we really want to measure. Consider the prediction problem of deciding whether a student will be successful (e.g., in a college admissions context). Fully capturing the outcome of 'successful student' in terms of a single measurable attribute is impossible because of its complexity. In cases such as these, algorithm designers resort to some available label such as 'GPA' (Kleinberg et al. 2018), which ignores different indicators of success achieved by parts of the population.

1. **The measurement process varies across groups.** For example, if a group of factory workers is more stringently or frequently monitored, more errors will be observed in that group. This can also lead to a feedback loop wherein the group is subject to further monitoring because of the apparent higher rate of mistakes (Barocas and Selbst 2016).

2. **The quality of data varies across groups.** Structural discrimination can lead to systematically higher error rates in a certain group. For example, women are more likely to be misdiagnosed or not diagnosed for conditions where self-reported pain is a symptom (Calderone 1990). In this case, "diagnosed with condition X" is a biased proxy for "has condition X."

# Historical bias

**Historical bias** arises when there is a misalignment between world as it is and the values or objectives to be encoded and propagated in a model. It is a normative concern with the state of the world, and exists even given perfect sampling and feature selection.

1

**Example: image search** In 2018, 5% of Fortune 500 CEOs were women (Zarya, 2018). Should image search results for “CEO” reflect that number? Ultimately, a variety of stakeholders, including affected members of society, should evaluate the particular harms that this result could cause and make a judgment. This decision may be at odds with the available data even if that data is a perfect reflection of the world. Indeed, Google has recently changed their Image Search results for “CEO” to display a higher proportion of women.

# Fairness in algorithms

- Increasing attention on algorithms being **fair**, not just accurate
- Fairness can be measured as:
  - demographic (or statistical) parity: population percentage should be reflected in the output classes
  - Equality of false negatives or equalized odds: constant false-negative (or both false-negative and true-negative) rates across groups.
  - Equal opportunity: equal True Positive Rate for all groups
  - Other metrics...
- Accuracy and fairness tend to be at odds with each other.
- Algorithms can be audited to test their fairness.
- *Are we ethically required to sacrifice accuracy for fairness? To what extent?*

# When the metric becomes the target (Goodhart's Law)

*"When a measure becomes a target it ceases to be a good measure"*

- Metrics introduced in the [British public healthcare system](#) (e.g. waiting time in ER) caused people to game it:
  - Cancelled scheduled operations to draft extra staff to ER
  - Required patients to wait outside the ER, e.g. in ambulances
  - Put stretchers in hallways and classified them as "beds"
  - Hospital and patients reported different wait times
- Big Data is significantly changing college applications
  - Universities are given higher ranking for things such as receiving more applications, being more selective, and having more students accept their offers (while tuition is not considered)
  - This even pushed some mid-tier universities to reduce the number of offer letter sent out, especially to good students who they think will not accept. Can affect applications to "safety schools"
- *Can you think of more examples?*
- *Can you think of ways to avoid this trap?*

# Ethics of pricing algorithms

- Algorithms are currently used to adjust prices based on:
  - Willingness of buyer
  - Availability
- Uber surge pricing:
  - In 2014, terrorists attacked a café in Sidney, holding 10 customers and 8 employees hostage for 16 hours
  - During this time, people from the surrounding areas were evacuated. Transportation was disrupted.
  - Uber prices adapted by increasing the rate to a minimum of 100\$
  - In general, underserved (poorer) areas get worse rates under current pricing policy
  - Drivers are also subjected to different pricing/waiting times
    - [https://papers.ssrn.com/sol3/papers.cfm?abstract\\_id=4331080](https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4331080)
- *Is Uber morally obliged to avoid such pricing disparities?*

# From trading to gambling addiction?

- Case study: Robinhood Trading app
- Designed to make trading more accessible and equitable. Robinhood's mission statement is "To democratize finance for all"
- Concerns around the most “dopamine-inducing” features ([source](#)):
  - Green confetti to celebrate transactions.
  - A constant update of stock related articles.
  - A colorful, eye-catching interface.
  - Emoji phone notifications.
  - One-click trading for instant gratification.
  - Free stocks in the shape of lottery tickets.



# Technology that will be in widespread use

- Tailored solutions for **specific tasks**, not general intelligence
- Prototypes **that work today** in labs & narrow deployments
- Some examples:
  - **Non-text input modalities** (vision; speech)
  - **Consumer modeling** (recommendation; marketing)
  - **Cloud services** (translation; question answering; AI-mediated outsourcing)
  - **Transportation** (automated trucking; some self-driving cars)
  - **Industrial robotics** (factories; some drone applications)
  - **AI knowledge work** (logistics planning; radiology; legal research; call centers)
  - **Policing & security** (electronic fraud; cameras; predictive policing)



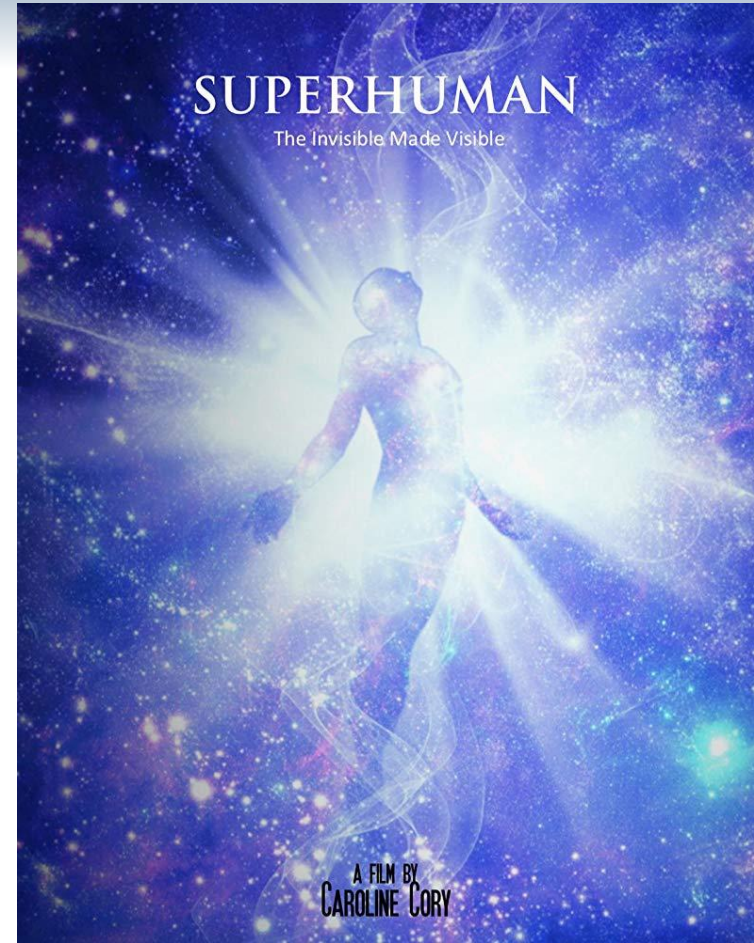
# Technologies that won't take off as quickly

- Overall, areas in which
  - major entrenched **regulatory regimes** need to be navigated
  - there exist **social/cultural barriers** to adoption
  - the **human touch** is crucial
  - substantial **new hardware** would need to be developed
- Some **examples**:
  - childcare, healthcare, eldercare
  - education
  - coaching, counselling
  - consumer robots beyond niche applications
  - semantically rich language understanding



# Superhuman Intelligence

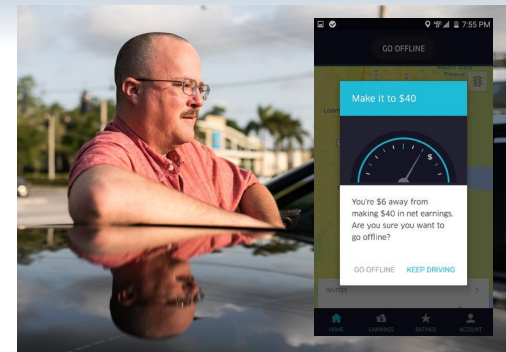
- AI systems will increasingly be capable of reaching **human-level performance**
- **Superhuman intelligence** isn't such a foreign, scary thing
  - governments, corporations, NGOs exhibit behavior much more sophisticated and complex than that of any individual
- Many important problems need superhuman intelligence; AI can help
  - improved **collective decision making**
  - more efficient use of **scarce resources**
  - addressing **underserved communities**
  - **climate change**; other societal challenges



# Ethics of AI

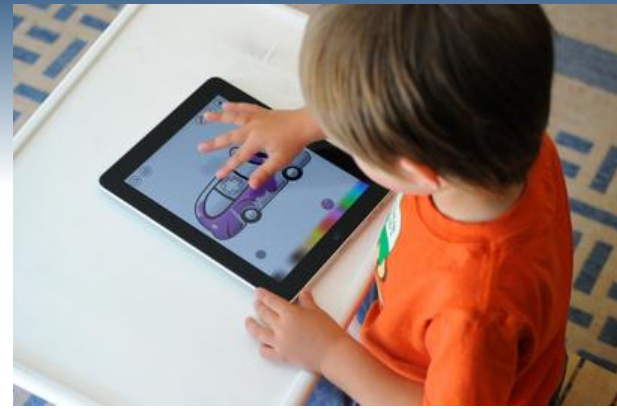
## Will a new technology:

- disempower **individuals vs corporations?**
  - ⇒ user modeling; **data mining**; fostering addictive behaviors; developmental effects on children
- disempower **individuals vs governments?**
  - ⇒ facilitate disinformation (deep fakes; bots masquerading as people; filter bubbles); enable qualitatively new military or security tactics
- take **autonomous actions** in a way that obscures responsibility
  - ⇒ autonomous weapons; self-driving cars; loan approval systems
- disproportionately affect **vulnerable/marginalized groups**
  - ⇒ automated decision making tools trained in ways that may encode existing biases



# Social Impact

- How will AI technologies **transform society**?
- Will there be a **social backlash** against AI?
  - If so, what will be considered AI?
- This **generation of children** will grow up taking for granted many technologies that strike us as magical
- How will **human relationships** change in the presence of always-available social agents?
- As we are increasingly **augmented by AI**, what are our inherent cognitive/emotional/motivational limitations, beyond which augmentation won't help?



# Ethical consideration of advances in AI

- *Is it wrong to create machines capable of making human labor obsolete? Will humans become demoralized by the presence of vastly more intelligent robots?*
- *How can we ensure that intelligent robots will not be put to an evil purpose by a malevolent human? How can we ensure they do not adopt malevolent purposes themselves?*
- *Is it morally acceptable to create “personal” (self-conscious) AI?*

